PATIENT-SPECIFIC COMPUTATIONAL MODELING OF AORTIC DUAL CIRCULATION IN HARLEQUIN SYNDROME DURING PERIPHERAL VA-ECMO

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Introduction

Peripheral VA-ECMO creates competing retrograde extracorporeal and antegrade native flows, producing aortic dual circulation that may lead to differential hypoxaemia (Harlequin syndrome) [1]. Their interaction determines both the position of the source interface and the origin of blood supplied to individual supra-aortic branches. Global pressure and flow metrics cannot resolve this spatial source distribution. We therefore developed a source-resolved numerical framework to characterize native-ECMO flow competition and the aortic blood-source distribution relevant to Harlequin syndrome.

Methods

A lumped-parameter cardiovascular model configured for peripheral VA-ECMO generated two internally consistent operating states spanning native-dominant and ECMO-dominant support. State-specific pulsatile left-ventricular (LV) outflow and effective retrograde distal flow were transferred one-way to a thoracic-aortic CFD model including the brachiocephalic artery (BCA), left common carotid artery (LCCA), and left subclavian artery (LSA). The distal contribution represented flow reaching the truncated thoracic domain and was not equated to total pump flow. RCR Windkessel models represented downstream impedances. Following solution of the incompressible Navier-Stokes equations, a passive scalar was transported by advection-diffusion, with c=1 marking LV-derived and c=0 ECMO-derived blood. Branch source fractions were computed from positive outgoing flow-weighted fluxes. The interaction region was defined by $0.25 \leq c \leq 0.75$, while c=0.5 identified a composition interface rather than a stagnation surface. Mesh sensitivity and cycle-to-cycle convergence were assessed. The workflow separates bulk haemodynamics from blood-source composition [2].

Results

In the native-dominant state, ECMO-derived fractions at the BCA, LCCA, and LSA were 6.9%, 10.4%, and 75.4%, respectively; in the ECMO-dominant state, they increased to 53.3%, 78.4%, and 99.9%. The interaction region shifted proximally under ECMO-dominant support. Although relative branch-flow distribution changed only marginally, source composition changed markedly, showing that source-resolved metrics provide information not identifiable from bulk flow alone.

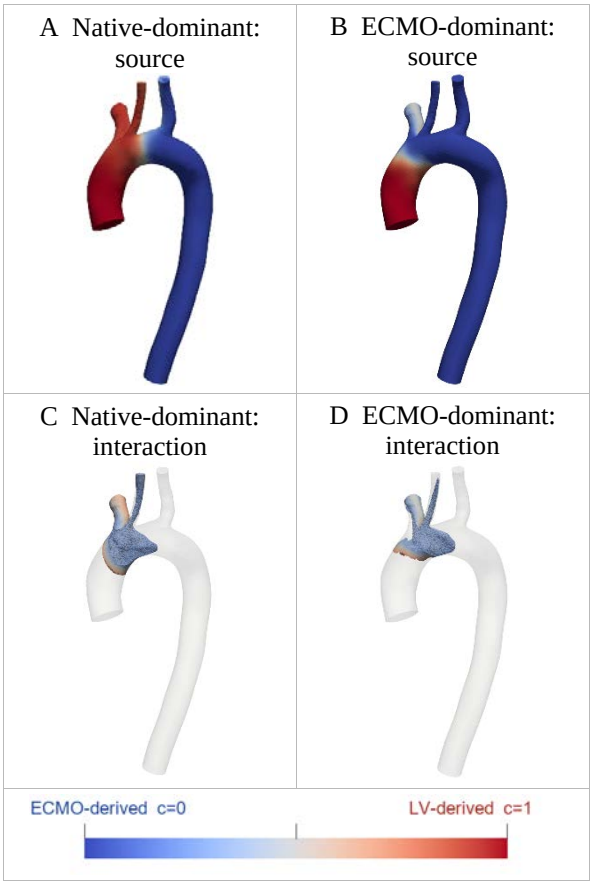


Figure 1. Proof-of-concept source composition (A-B) and interaction regions (C-D; $0.25 \leq c \leq 0.75$) for native-dominant and ECMO-dominant operating states.

Conclusions

The framework links systemic operating states to three-dimensional aortic flow and quantifies blood-source composition independently of branch flow, providing a haemodynamic basis for studying Harlequin syndrome. The scalar marks blood source, not oxygen content, and the distal condition is an imposed effective flow. Future work will include bidirectional distal flow, oxygen transport, closed-loop 0D-3D coupling, patient-specific inputs, and validation.

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References

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